FREIGHT RATE PREDICTION

Machine Learning Engineering Assessment

Technical Report

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Date: November 11, 2026

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1. Project Overview

In this project I built a regression model that estimates the posted freight rate for a shipment. I was given a labeled development dataset with historical loads, and a separate validation dataset of 12,000 loads that needed predictions. I also had to produce a daily prediction for one fixed shipment across every day of December 2025.

The target variable is posted_rate. I prepared the final prediction files in the formats the assessment asked for and checked them with the scoring script that came with it.

2. Dataset Overview

The development dataset has 48,000 rows and 14 columns. The validation dataset has 12,000 rows and 13 columns. The December scenario file has 31 rows and seven input columns.

Dataset	Rows	What I used it for
Development data	48,000	Training the model and checking it internally
Validation data	12,000	Final load rate predictions
December inputs	31	Daily predictions for the fixed shipment

The development data runs from January 1 to October 31, 2025, and the validation data runs from November 1 to December 31, 2025. The December scenario is a Dry Van load of 32,000 lb going 360 miles from Lexington (pickup) to Fort Wayne (delivery).

3. Exploratory Data Analysis

Before modeling, I looked at how the freight rates are distributed, how rate relates to distance, and how the main categorical variables behave.

3.1 Target variable

The posted_rate target has 48,000 observations. The mean is about $2,373.98 and the median is about $2,030.76, with a standard deviation of about $1,486.49. Rates range from $57.22 up to $25,533.

3.2 Distance and freight rate

Distance and posted_rate are strongly and positively correlated (about 0.9085), so distance is clearly one of the most useful predictors of cost. Weight is only weakly correlated with the target (about 0.0348), but I kept it in as a feature in case it helps in combination with other fields.

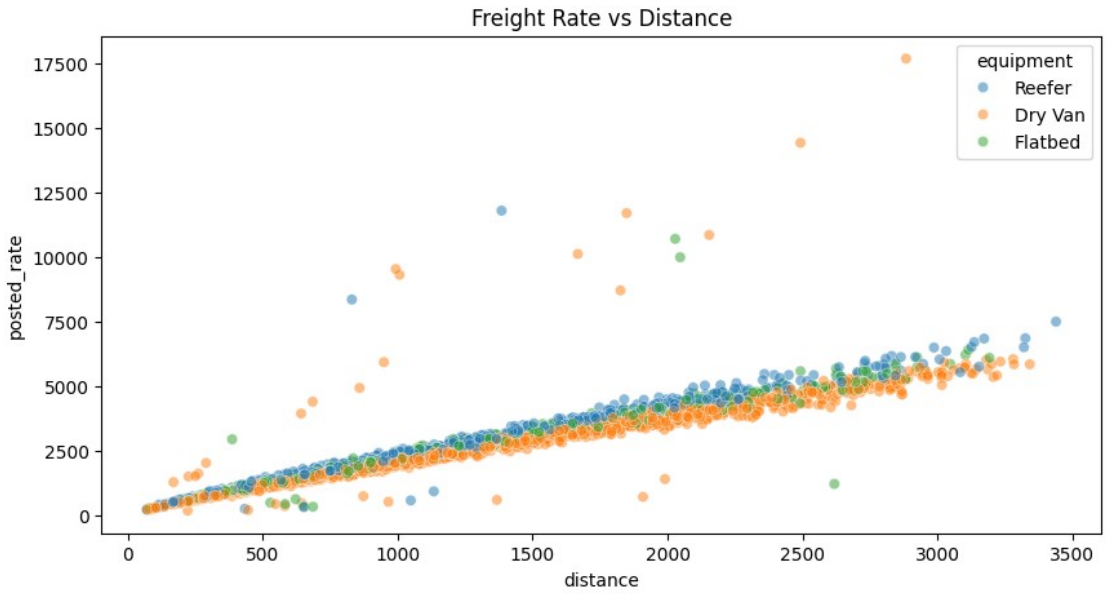


Figure 1. Freight rate vs. distance, coloured by equipment type

Most loads sit tightly along a rising line, and there are a handful of very high or very low outliers that do not follow it.

3.3 Freight distribution

The development data has 27,202 Dry Van rows, 12,045 Reefer rows and 8,753 Flatbed rows. Average posted rates differ a bit by equipment: about $2,271.55 for Dry Van, $2,553.64 for Reefer and $2,445.09 for Flatbed.

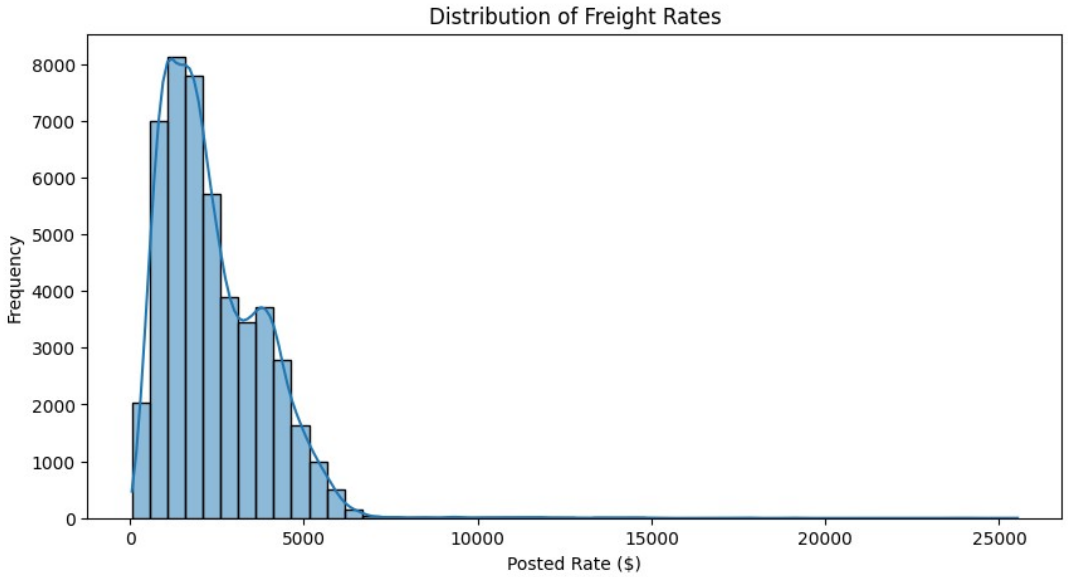


Figure 2. Distribution of freight rates

The distribution is skewed to the right. Most rates fall between a few hundred and about $6,000, with a very thin tail of expensive loads.

4. Data Quality and Preprocessing

The development data has missing values in market_index (374 rows) and weight (300 rows). The validation data is missing market_index in 249 rows and weight in 165 rows. In the pipeline I filled numeric gaps with the

median and categorical gaps with the most frequent value, and then one-hot encoded the categorical variables.

Dates were converted to datetime and split into month, day of month and day of week. After that I dropped the raw date column from the model input. I did not use load_id as a feature.

The December input file does not have the geographic coordinates, market_index or quote_signal fields that the development data has. So that the same feature set could be used for both final prediction files, the final model only uses the fields all three datasets share: pickup, delivery, distance, equipment, weight and the date-based features.

5. Validation Strategy and Model Development

I used a chronological split for internal validation because the data is time based. The model was trained on January to September 2025 and tested on October 2025, so later observations never leak into the training period.

I tried several HistGradientBoostingRegressor settings. This is a tree-based gradient boosting model that can pick up non-linear patterns and interactions between shipment features. I also tested a log1p transformation of the target, converting predictions back to the normal rate scale with expm1.

In the October experiments, a reduced-feature log-target model reached an MAE of about $112.65, an RMSE of about $648.61 and an R-squared of about 0.8199. These numbers come from that experiment only. The final shared-feature model was trained afterwards on all 48,000 development rows so it could predict both the validation and December data with the same inputs, so the October metrics should not be read as the measured performance of the final model.

6. Final Model and Prediction Generation

The final model is a HistGradientBoostingRegressor with 200 boosting iterations, a learning rate of 0.07, at most 31 leaf nodes, L2 regularization of 2 and random_state 42. It was trained on the log1p scale and its predictions were converted back with expm1. I also set a small positive lower bound on the predictions, since the assessment requires positive rates.

- Validation predictions were generated for all 12,000 rows.
- The validation output has exactly two columns: load_id and predicted_rate.
- December predictions were generated for all 31 dates.
- The December output has pickup, delivery, distance, equipment, weight, date and predicted_rate.

7. December Scenario Results

For the December scenario the shipment details stay fixed and only the date changes. The scorer validated all 31 daily records and produced the chart below.

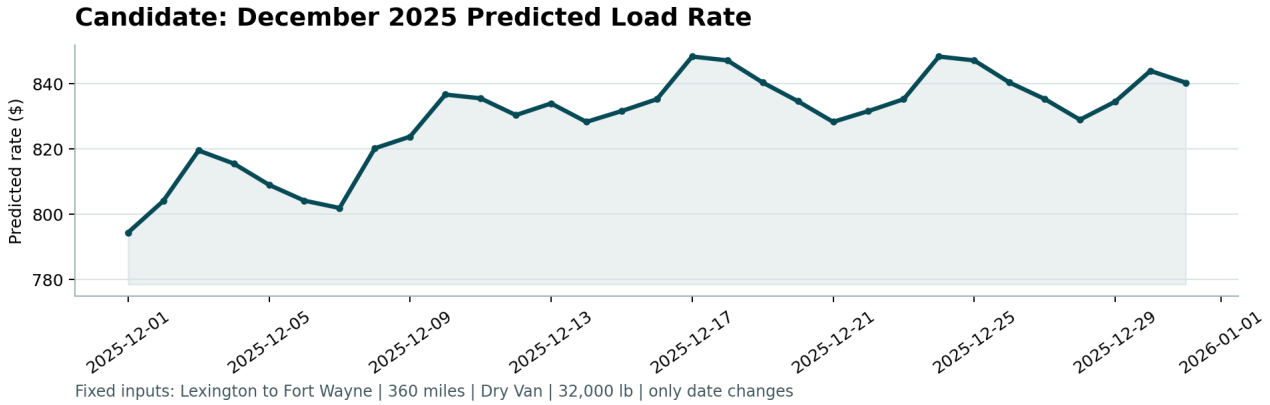


Figure 3. Predicted load rate for each day of December 2025

The predicted rate starts near $795 at the beginning of the month and moves up to the mid $840s by mid and late December, with small ups and downs along the way.

8. Scorer Verification

The scorer provided with the assessment ran successfully and returned these messages:

- Validated 12,000 final predictions.
- Validated 31 fixed December predictions.
- Created chart: scorer_results\candidate_december.png

The scorer only checks that the prediction files are structured correctly and creates the December chart. It does not calculate the final validation metrics. Spotter calculates those after submission.

9. Conclusion

In this project I went through the full workflow for a freight rate regression problem. I explored the historical data, dealt with missing values, created date features, compared several model experiments, and produced predictions for the validation data and the fixed December scenario. Both prediction files passed the scorer's structure and scenario checks. How well the model does on the hidden validation targets will only be known once the assessment platform scores it after submission.

10. Reproducibility and Submission Files

The submission repository should contain:

- The finished Jupyter notebook with the data exploration, preprocessing, training and prediction code.
- score.py and requirements.txt.
- validation_predictions.csv.
- The December prediction input/output file that the README and scorer ask for.
- scorer_results/candidate_december.png.
- This technical report, plus clear run instructions in README.md.

Before publishing the repository, I will double check that the file paths in the README match the real folder structure. The scorer command in the README uses the --predictions and --december-predictions arguments.